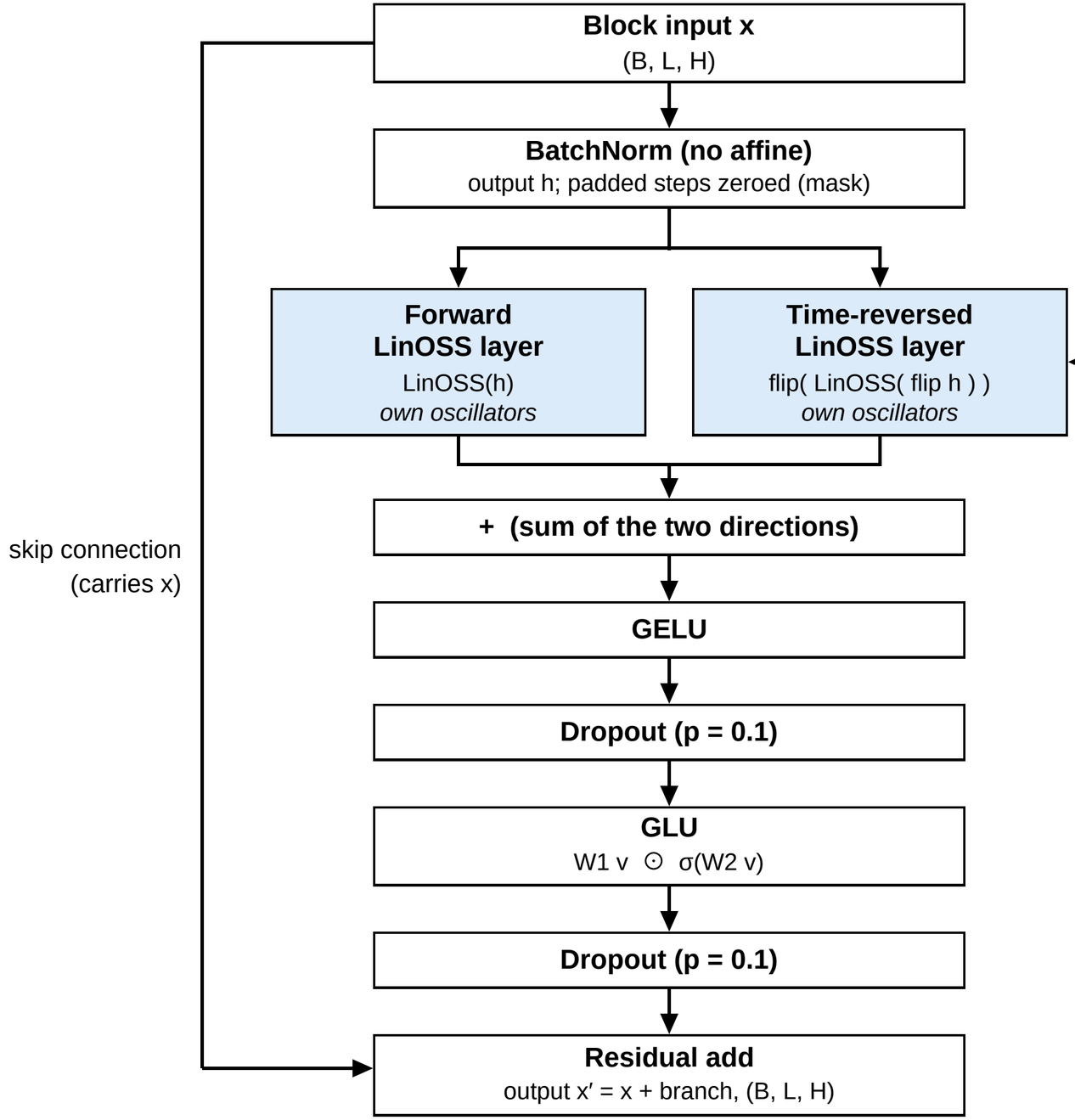



**Each LinOSS layer (both directions)**

$$y'' = -A\,y + B\,u$$

$$x = \mathrm{Re}(C\,y) + D\,u$$

A = relu(A_raw) ≥ 0 (diagonal); B, C complex

u: layer input; x: layer output

step dt = sigmoid(steps_raw) per oscillator

implicit (IM) discretisation: always stable

**Natural periods initialised log-uniform 5–240 s**
(at one token per second)

**Block sizes in OscSeq**

1 block per electrode: H = 32, P = 32

*(weights shared over electrodes)*

2 blocks over the whole head: H = 96, P = 96

P: oscillators per direction; H: channels